



# **Models and their Management**

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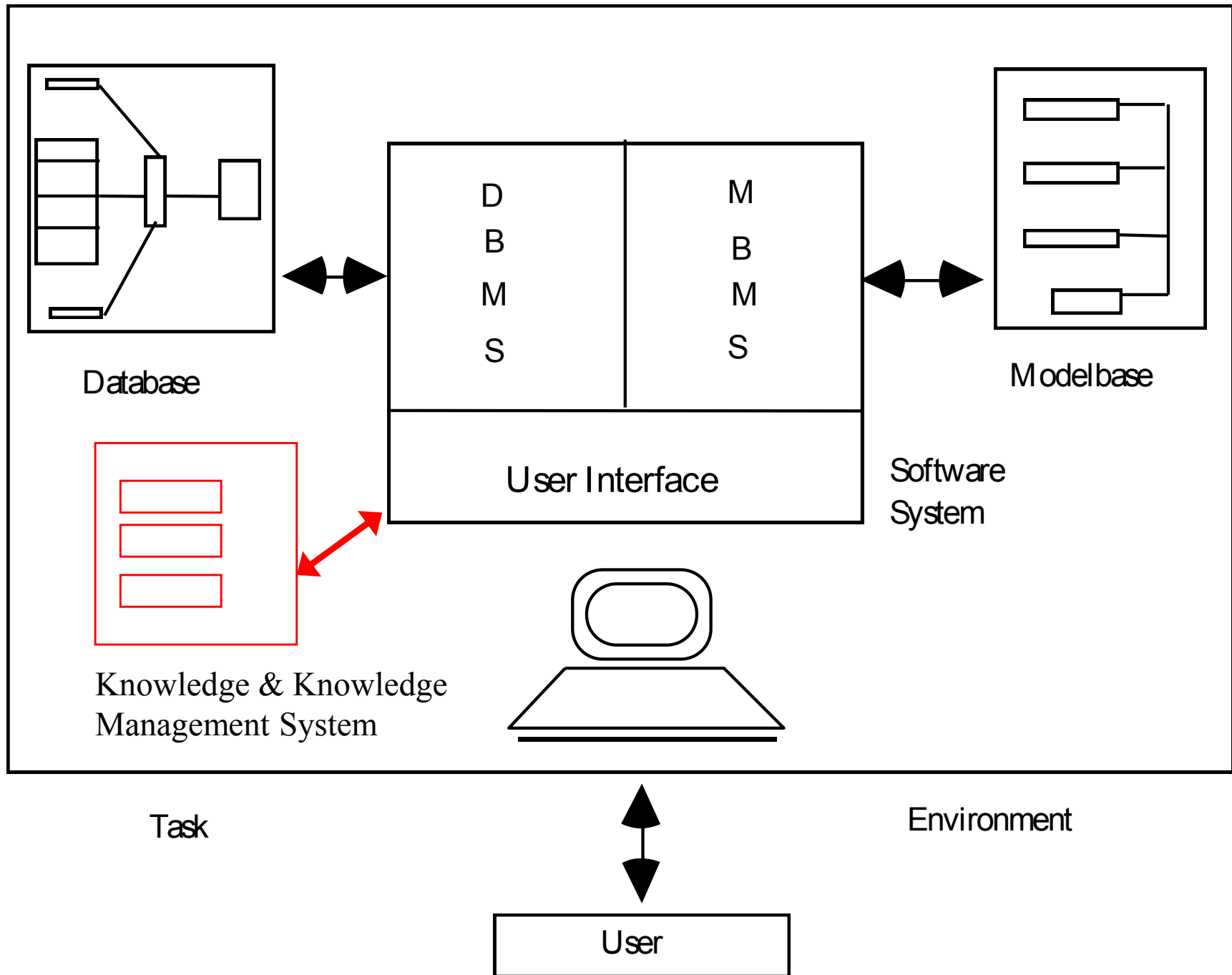
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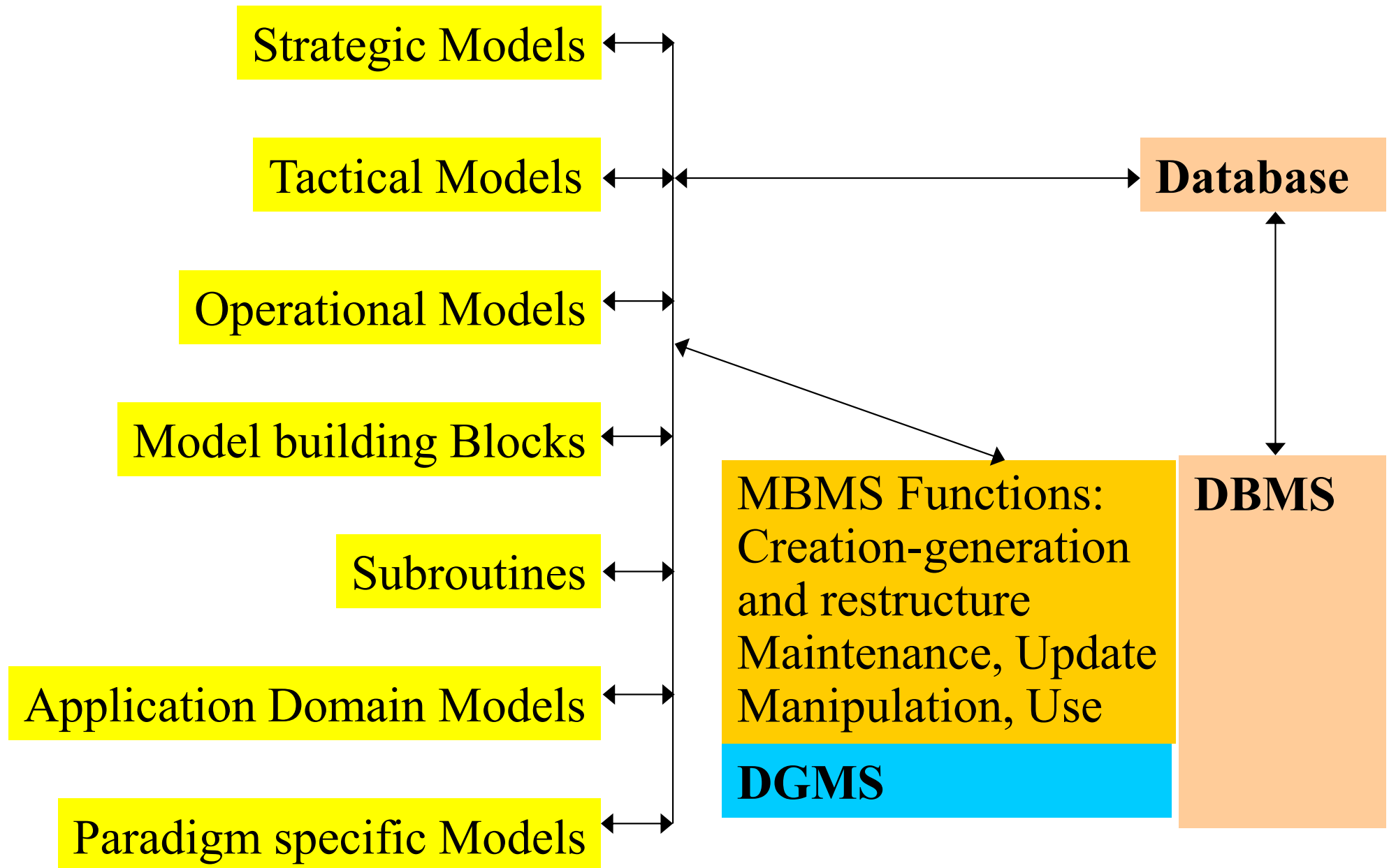
# Objectives

- ★ DSS Components
- ★ Modelbase Management System
- ★ Models
- ★ Representation of Models
- ★ Process of Modelling
- ★ Integration of Models

# Sprague & Carlson



# Modelbase Management Systems



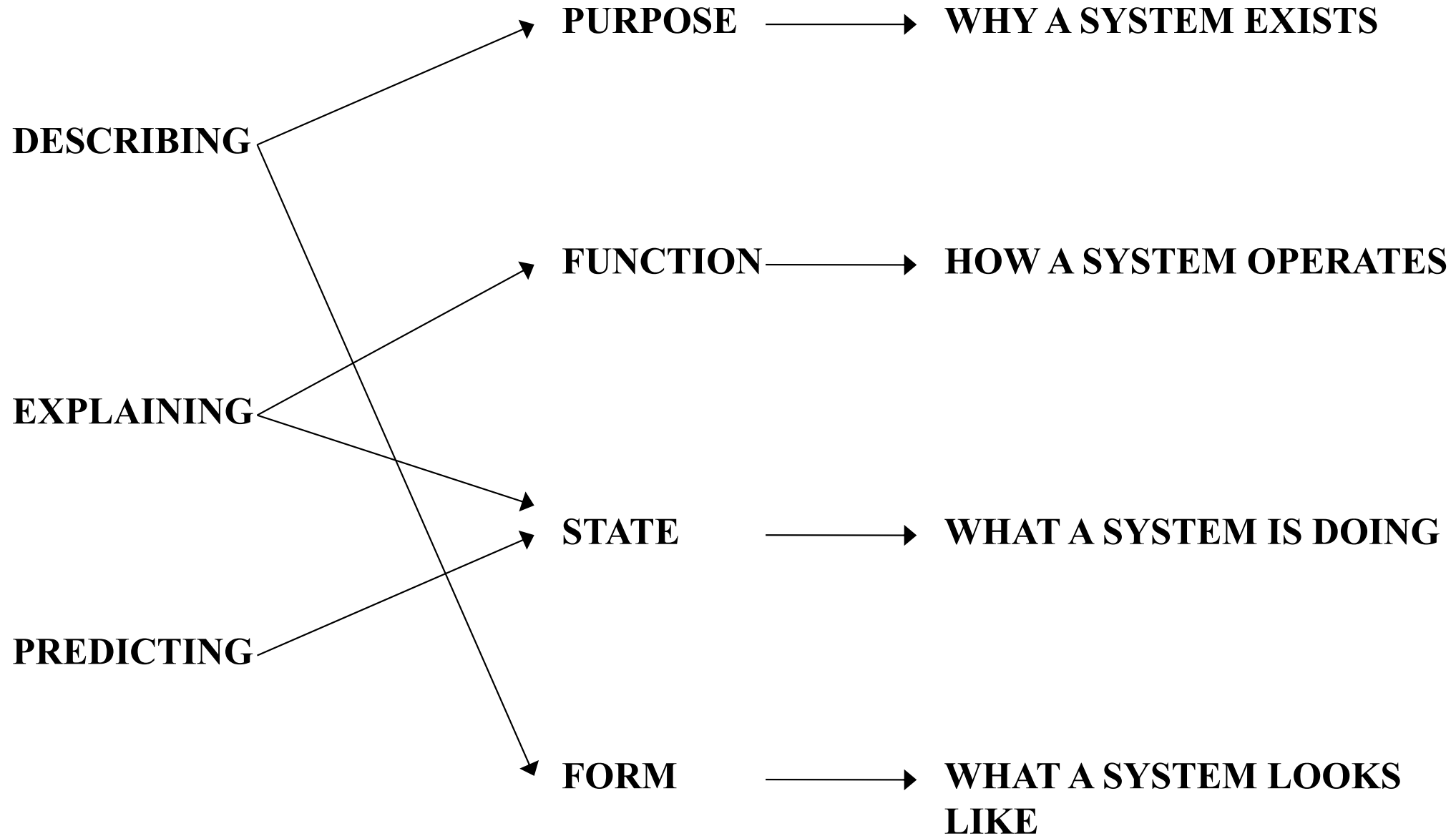
# Models

- ★ Models are central to DSS - perform analysis on a model of reality than on the real system itself
- ★ A model is a simplified abstraction of reality
- ★ Types of models (least abstract to most abstract)
  - Iconic (3D, photographs, GUI, OO,etc.)
  - Analog (blueprints, orgn. and stock market charts, etc.)
  - Mathematical/Quantitative Models

## ★ Benefits of Models

compression of time, easier and less expensive to experiment, making mistakes not costly, can incorporate complexity and uncertainty, explore huge number of alternatives, enhance and reinforce learning and training

# Purpose of Models



# Models and Managers - J.D.C. Little

- ★ “The big problem with management science models is that managers practically never use them.”
- ★ Some of the reasons for this:
  - Good models are hard to find
  - Good parameterisation is even harder
  - Difficulty of understanding complex models
- ★ To overcome this decision models should be:
  - simple, robust, and adaptive
  - easy to control
  - complete on important issues and
  - easy to communicate with

# Categories of Models

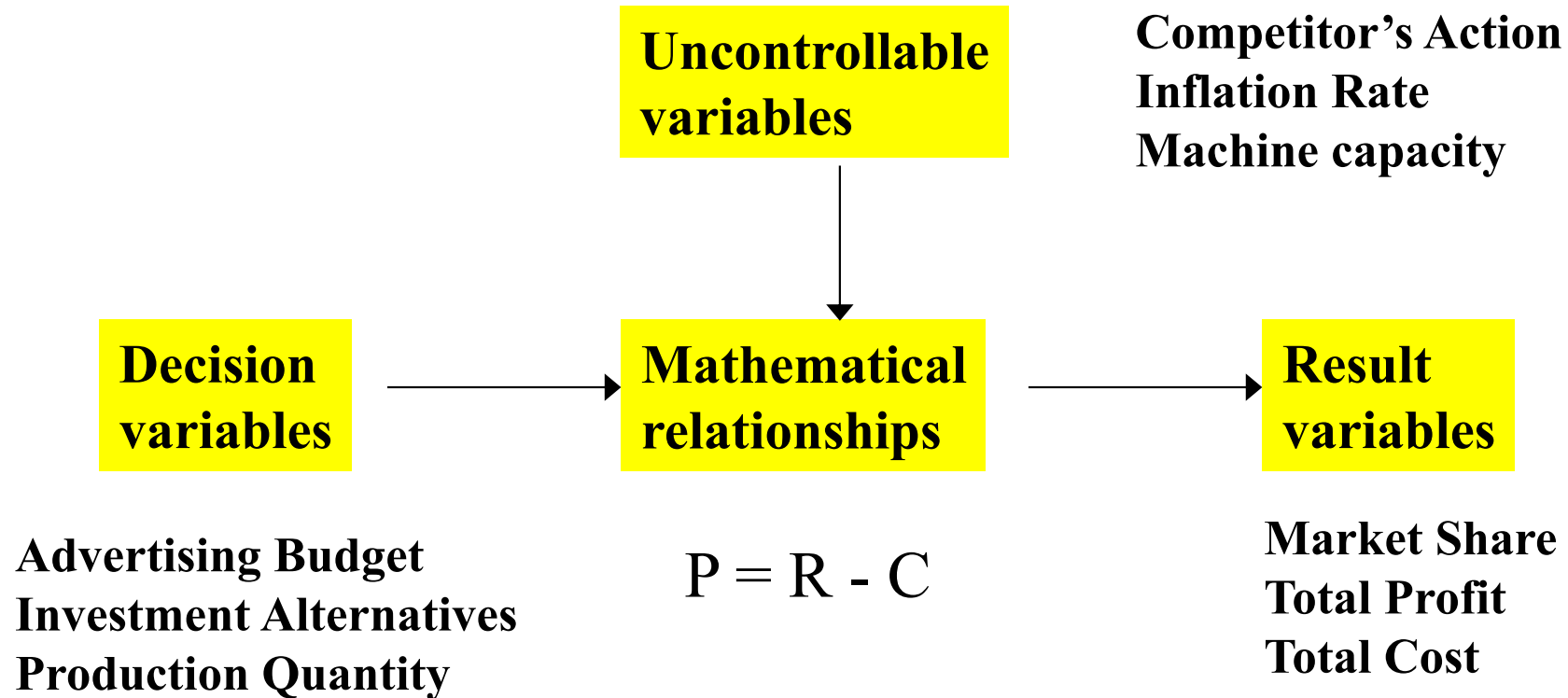
| Category                                      | Process and Objective                                                                               | Techniques                                                       |
|-----------------------------------------------|-----------------------------------------------------------------------------------------------------|------------------------------------------------------------------|
| Optmization of problems with few alternatives | Find the best solution from a small number of alternatives                                          | Decision Tables and Decision Trees                               |
| Optimization using an algorithm               | Find the best solution from a large number of alternatives using a step-by-step improvement process | Linear and other mathematical programming models, network models |
| Optimization via analytical formula           | Find the best solution, in one step, using a formula                                                | Some inventory models                                            |
| Simulation                                    | Finding a good enough solution, or the best among the alternatives checked, using experimentation   | Several types of simulation                                      |
| Heuristics                                    | Find a good enough solution using rules                                                             | Heuristic programming, expert systems                            |
| Predictive models                             | Predict future for a given scenario                                                                 | Forecasting models, Markov analysis                              |
| Other models                                  | Finding What-if using a formula                                                                     | Finanicial modelling, waiting lines                              |



# Characteristics of Models

- ★ Static or dynamic
- ★ Normative, Descriptive, and Prescriptive
- ★ Sub-optimisation and Satisficing
- ★ Certainty, uncertainty, or risk

# General Structure of a Quantitative Model



- **Data** to instantiate Model
- **Solver** to execute/solve the Model
- **Visualisation** to visualise results obtained from execution/solving of Model

# **Representation of Models**

- ★ Richness of the representation
- ★ Levels of Abstraction
- ★ Paradigm and Domain Independence
- ★ Model Reusability

# No Model/Data Independence

$$\max \quad Z = 3x + 5y,$$

$$\text{s.t.} \quad 5x + 2y \leq 12,$$

$$7x - 3y \leq 15,$$

$$x, y \geq 0.$$

# Value Independence Only

$$\begin{array}{ll}\max & Z = c_1x_1 + c_2x_2, \\ \text{s.t} & a_{11}x_1 + a_{12}x_2 \leq b_1, \\ & a_{21}x_1 + a_{22}x_2 \leq b_2, \\ & x_1, x_2 \geq 0,\end{array}$$

$$\mathbf{c} = \begin{pmatrix} 3 \\ 5 \end{pmatrix}, \quad \mathbf{a} = \begin{pmatrix} 5 & 2 \\ 7 & -3 \end{pmatrix}, \quad \mathbf{b} = \begin{pmatrix} 12 \\ 15 \end{pmatrix}, \quad \mathbf{x} = \begin{pmatrix} ? \\ ? \end{pmatrix}.$$

# Dimension & Value Independence

$$\begin{aligned} \max \quad & \sum_{j=1}^n c_j x_j, \\ \text{s.t.} \quad & \sum_{j=1}^n a_{ij} x_j \leq b_i, \quad i = 1, \dots, m, \\ & x_1, x_2, \dots, x_n \geq 0, \\ & \mathbf{c} = \begin{pmatrix} 3 \\ 5 \end{pmatrix}, \quad \mathbf{a} = \begin{pmatrix} 5 & 2 \\ 7 & -3 \end{pmatrix}, \quad \mathbf{b} = \begin{pmatrix} 12 \\ 15 \end{pmatrix}, \quad \mathbf{x} = \begin{pmatrix} ? \\ ? \end{pmatrix}. \end{aligned}$$

# Data Structure Independence

(1) Store each vector and matrix in a separate file.

| <i>c</i> | Column 1 | Column 2 | <i>b</i> |
|----------|----------|----------|----------|
| 3        | 5        | 2        | 12       |
| 5        | 7        | -3       | 15       |

(2) Store all vectors and matrices in a single file.

| Vector   | Column 1 | Column 2 |
|----------|----------|----------|
| <i>c</i> | 3        |          |
| <i>c</i> | 5        |          |
| <i>a</i> | 5        | 2        |
| <i>a</i> | 7        | -3       |
| <i>b</i> | 12       |          |
| <i>b</i> | 15       |          |

# Data Structure Independence

3. Store vectors *c* and *b* together in one file. Store matrix *a* in a separate file, one element per record with explicit subscripting.

| Vector | Value |
|--------|-------|
| c      | 3     |
| c      | 5     |
| b      | 12    |
| b      | 15    |

| Matrix a |        |       |
|----------|--------|-------|
| Row      | Column | Value |
| 1        | 1      | 5     |
| 1        | 2      | 2     |
| 2        | 1      | 7     |
| 2        | 2      | -3    |



# Model/Solver Independence

```
DOUBLE PRECISION P, F, I, DENOM
READ F, I, N
DENOM = (1 + I) * * N
P = F/DENOM
PRINT P
END
```

```
DOUBLE PRECISION P, F, I
100 READ (INPUT, *, END = 900) F, I, N
P = PRESVAL (F, I, N)
PRINT P
GO TO 100
900 STOP
END
```

```
DOUBLE PRECISION FUNCTION PRES-
VAL (F, I, N)
DOUBLE PRECISION F, I
PRESVAL = F/(1 + I) * * N
RETURN
END
```

# Model/Solver Independence

```
DATA;
```

```
INPUT Y X1 X2 X3 X4;
```

```
CARDS;
```

```
10 12 45 67 49
```

```
12 10 37 60 92
```

```
...
```

```
PROC STEPWISE;
```

```
MODEL Y = X1 X2 X3 X4 / forward;
```

# Model/Solver Independence

(variable, set, and parameter definitions)

...

EQUATIONS

COST

the objective function

SUPPLY(I)

supply from plant I

DEMAND(J)

demand at location J

$\text{COST} \dots Z = \sum_{I,J} C(I,J) * X(I,J)$

$\text{SUPPLY}(I) \dots \sum_J X(I,J) = A(I)$

$\text{DEMAND}(J) \dots \sum_I X(I,J) = B(J)$

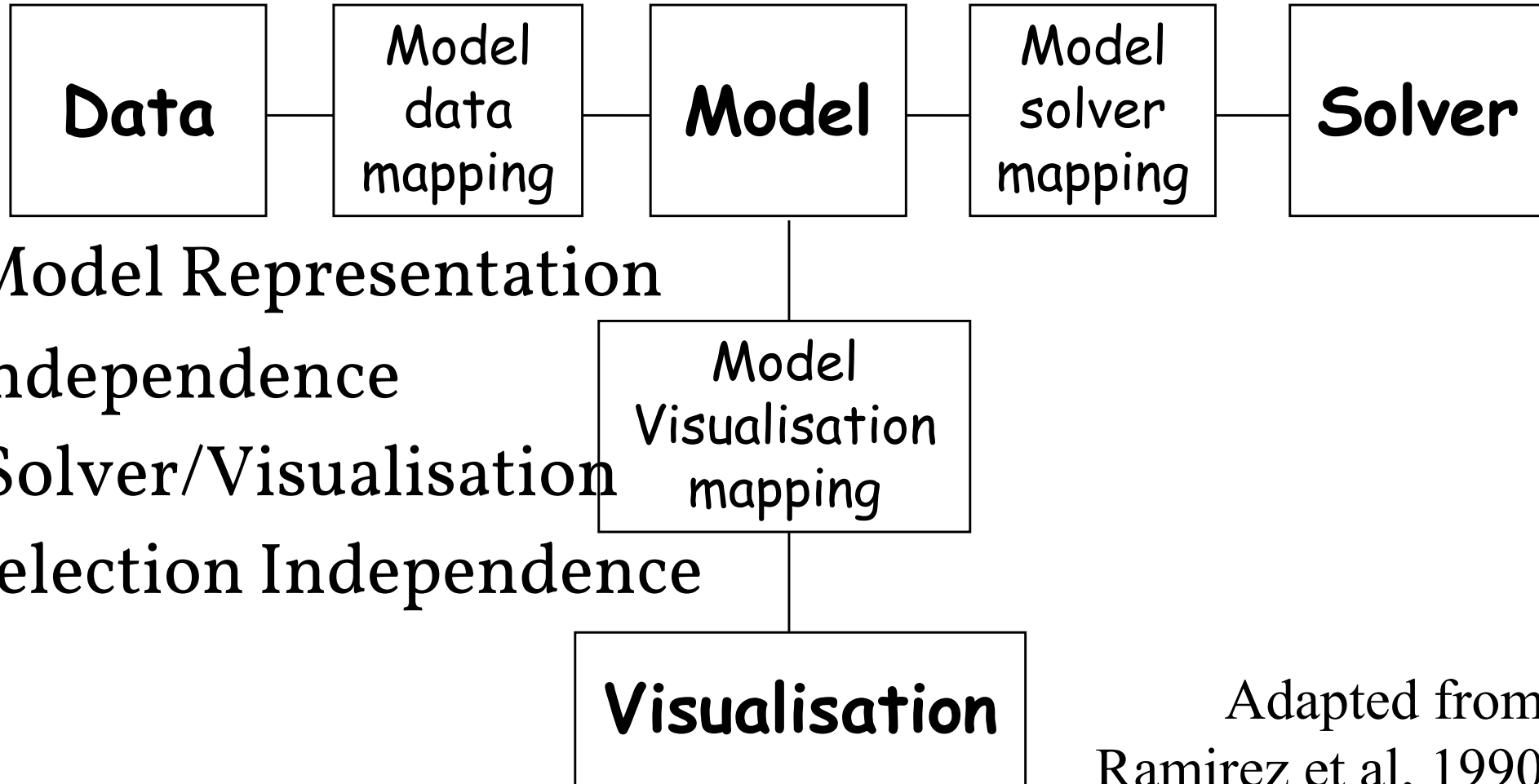
MODEL TRANSPORT /COST, SUPPLY, DEMAND

To solve this model, the user must issue the statement

SOLVE transport USING lp MINIMIZING z;

# Representation of Models

- ★ Model-Data, Model-Solver, Model-visualisation, Purpose, Direction Independence
- ★ Data Value & Dimension Independence



- ★ Model Representation Independence
- ★ Solver/Visualisation Selection Independence

Adapted from  
Ramirez et al, 1990

# **The Process of Modelling**

- ★ Problem Elicitation and Model Formulation
- ★ Model Instantiation - Data and Solver Integration
- ★ Model Storage - Structure and Instance
- ★ Model Termination
- ★ Model Integrity/Validity Checks
- ★ Model Execution and Reformulation Support
- ★ Model Versioning
- ★ What-if Analysis and Model Result Interpretation
- ★ Modelling System Interface
- ★ Model Usage Documentation
- ★ Model Sharing

# Integration of Models

## ★ Levels

- *Aggregation*
- *Pipelining and*
- *Splicing*

## ★ Types

- *Permanent and*
- *Ad-hoc*